Imbalanced Vehicle Classification

Phase 2 — Midterm Report

Introduction to Artificial Intelligence — Summer 2026

May 2026

1. Implementation Description

1.1 Architecture

We use ResNet-50 pretrained on ImageNet (torchvision) as our backbone. The final fully-connected layer is replaced with a linear layer mapping to 10 output classes. The full network is fine-tuned end-to-end, which proved necessary to adapt low-level features to aerial-view imagery.

1.2 Dataset Setup and Constraints

Due to hardware constraints (CPU-only training), we worked with a reduced sample of the full dataset to keep training times manageable. The dataset was loaded from the Kaggle archive structure (archive/train/EO) and split automatically using an 80/20 stratified split integrated in the training script — no separate val folder was needed.

To make CPU training feasible, the dataset was subsampled as follows:

- Training set: maximum 100 images per class → 1,000 images total
- Validation set: maximum 20 images per class → 200 images total

This subsampling naturally equalizes class sizes, which partially mitigates the imbalance problem at the dataset level. However, it also means results may not fully reflect performance on the complete 285,772-image training set.

1.3 Training Pipeline

The pipeline implements two combined strategies to address class imbalance:

- WeightedRandomSampler: each batch is sampled with probability inversely proportional to class frequency
- Weighted CrossEntropyLoss: misclassification of minority classes is penalized more strongly

Data augmentation applied during training: random horizontal/vertical flips, rotation ±15°, color jitter (brightness/contrast ±0.3), resize to 128×128, ImageNet normalization.

1.4 Hyperparameters

Parameter	Value
Input resolution	128 × 128
Batch size	64
Learning rate	1e-4
Optimizer	AdamW (weight decay 1e-4)
LR scheduler	Cosine Annealing
Loss function	Weighted CrossEntropyLoss
Train samples / class	100 (subsampled, CPU constraint)
Val samples / class	20 (subsampled, CPU constraint)
Backbone	ResNet-50 (pretrained ImageNet)
Device	CPU

2. Preliminary Results

2.1 Progression of Results across Epochs

We ran three experiments incrementally — 1, 5 and 10 epochs — to observe convergence behaviour. The table below summarises the key metrics extracted from the confusion matrices and training curves.

Experiment	Val Accuracy (approx.)	Macro F1 (approx.)	Main weakness
1 epoch (baseline)	~48%	~0.40	Sedan/SUV/Van heavily confused
5 epochs	~75%	~0.76	Sedan recall 30%, SUV 45%
10 epochs	~85%	~0.85	Sedan 60%, SUV 60% — still main errors

Macro F1 is the primary metric because the validation set is balanced — overall accuracy would be misleading if the model were biased toward majority classes.

2.2 Training Curves — 5 Epochs

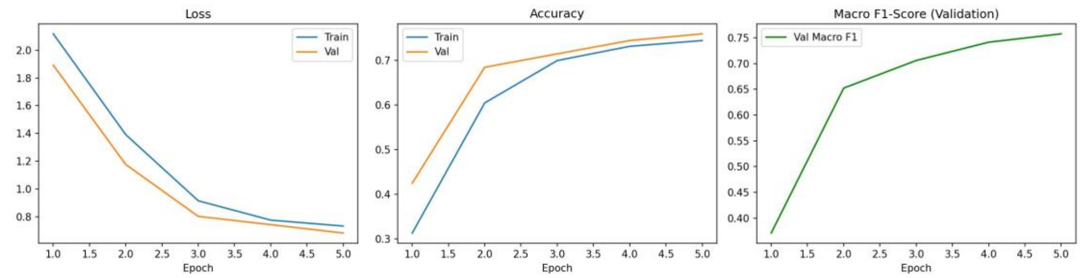

Figure 1 — Training and validation loss, accuracy, and Macro F1-score over 5 epochs

After 5 epochs, the loss drops steadily and the validation accuracy reaches ~75%. The Macro F1 score climbs from 0.40 to ~0.76, indicating that the model is learning to distinguish minority classes progressively.

2.3 Training Curves — 10 Epochs

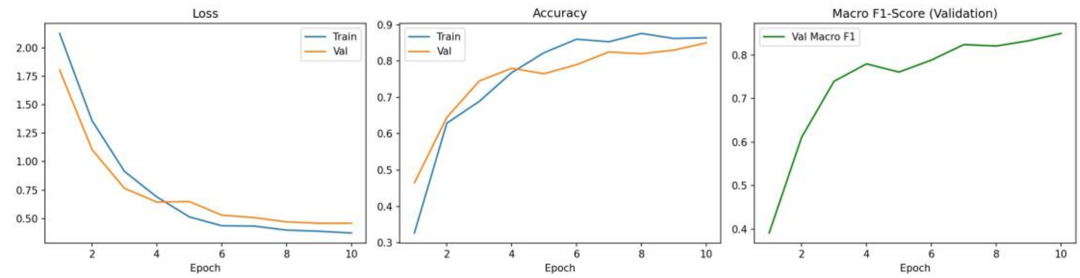

Figure 2 — Training and validation loss, accuracy, and Macro F1-score over 10 epochs

Extending training to 10 epochs shows continued improvement. The Macro F1 reaches ~0.85 and the validation accuracy ~85%. Importantly, the validation loss remains close to the training loss, suggesting no significant overfitting on the subsampled dataset.

2.4 Confusion Matrix — 1 Epoch (Baseline)

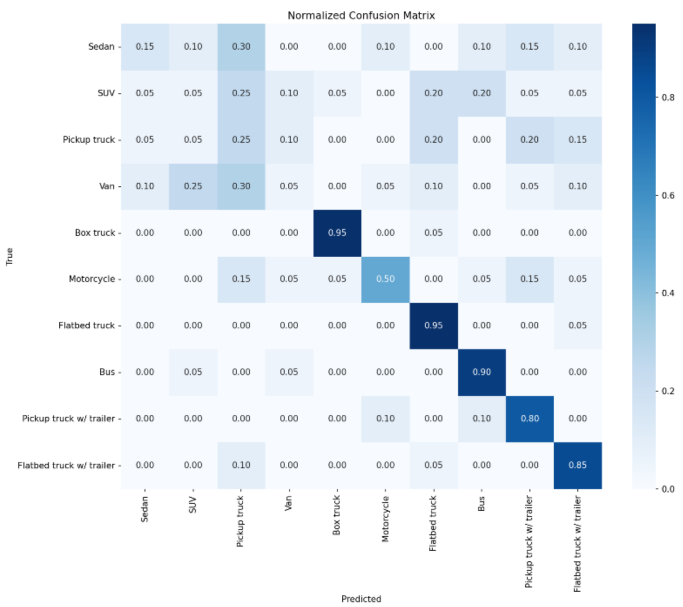

Figure 3 — Normalized confusion matrix after 1 epoch

At 1 epoch, the model already shows strong performance on visually distinct vehicle types (Box truck, Flatbed truck, Bus, trailers) with near-perfect recall. The main confusion zone is among passenger vehicles: Sedan is predicted correctly only 15% of the time, with frequent misclassification as Pickup truck (30%) and SUV (10%). Motorcycle reaches only 50% recall, and Pickup truck w/ trailer 80%. These baseline results confirm that transfer learning from ImageNet is immediately effective for vehicles with distinctive shapes, but insufficient for visually similar passenger vehicles seen from an aerial perspective.

2.5 Confusion Matrix — 5 Epochs

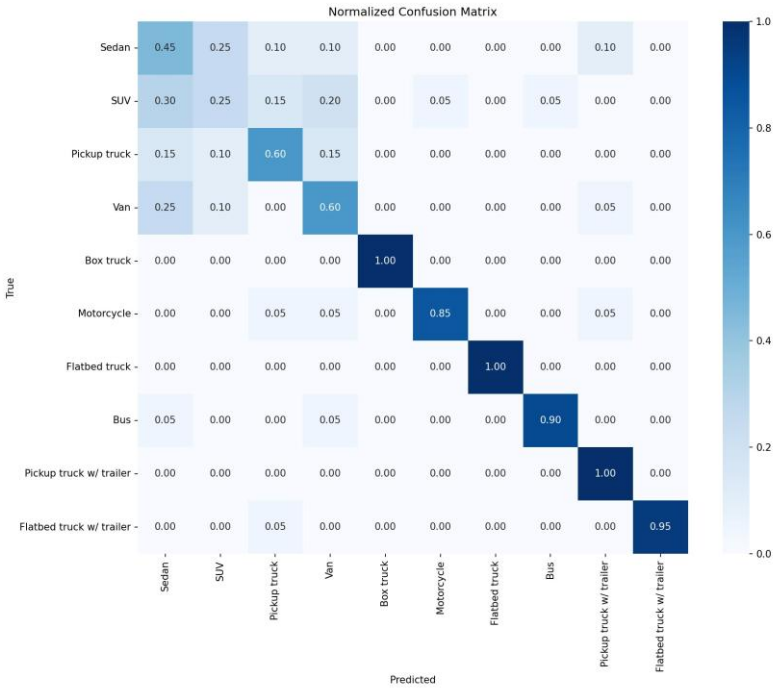

Figure 4 — Normalized confusion matrix after 5 epochs

After 5 epochs, performance on most classes is stable or slightly degraded compared to 1 epoch in some cases (e.g. Bus drops to 0.90, Flatbed truck w/ trailer to 0.95). This is likely due to the small validation set (20 images per class) creating some variance. Sedan improves slightly to 0.45 and Van to 0.60. The Sedan/SUV/Van confusion cluster remains the main challenge.

2.6 Confusion Matrix — 10 Epochs

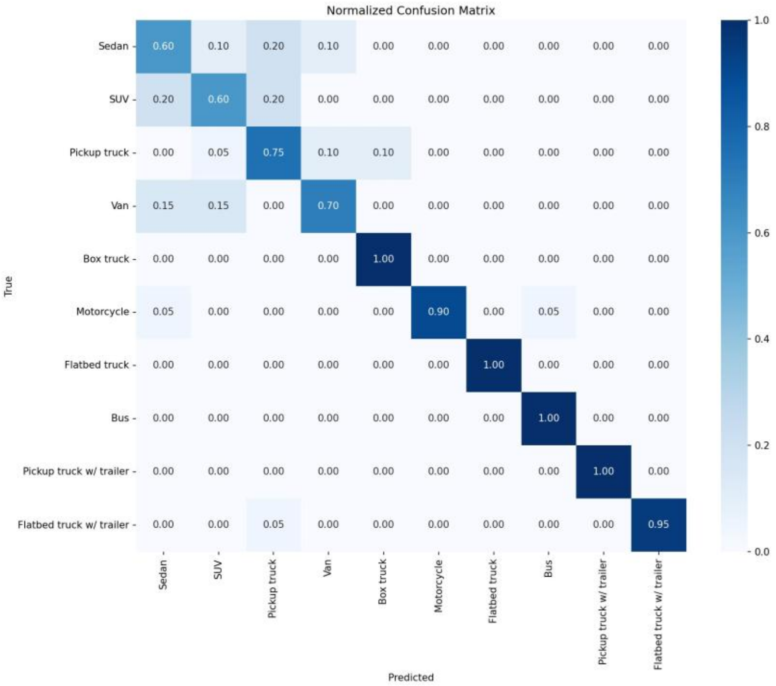

Figure 5 — Normalized confusion matrix after 10 epochs

At 10 epochs, clear improvement is visible across passenger vehicle classes. Sedan recall reaches 0.60, SUV 0.60, Pickup truck 0.75, and Van 0.70. Specialty vehicle classes (Bus, Box truck, Flatbed truck, trailers) maintain near-perfect recall. The main remaining confusion is between Sedan and Pickup truck (0.20 misclassification), which is expected given their visual similarity from an aerial perspective.

3. Experiment Analysis

3.1 What Worked

- Rapid convergence even on CPU with subsampled data: meaningful results were obtained from epoch 1, with consistent improvement through epoch 10.
- Specialty vehicle classes (Box truck, Flatbed truck, Bus, trailers) achieve near-perfect recall from the very first epoch, showing that transfer learning from ImageNet provides immediately useful features for these visually distinct shapes.
- Combined WeightedRandomSampler + weighted loss: the model avoids collapsing to a Sedan-predicting baseline, achieving balanced performance across all 10 classes.
- Cosine Annealing scheduler: smooth convergence observed — no instability or oscillation in the training curves across all three experiments.

3.2 What Did Not Work / Challenges

- CPU training speed: the primary bottleneck. With the full dataset (285,772 images), training is not feasible on CPU without subsampling. The 100 images/class limit is a necessary compromise that may affect generalisation.
- Sedan/SUV/Van confusion: these three classes remain the hardest to distinguish even at 10 epochs. From an aerial viewpoint, the difference in vehicle silhouette is subtle, and 100 training samples per class may be insufficient to learn discriminative features.
- Small validation set (20 images/class): creates noise in per-epoch metrics, visible in the slight dip in some class recall values between epoch 5 and 10 for rare classes.

3.3 Changes from Initial Plan

- Dataset structure: the Kaggle archive used a flat EO subfolder structure rather than separate train/val folders. We implemented an automatic 80/20 stratified split directly in the training script.
- Subsampling added: not in the initial plan — required due to CPU constraint. This effectively pre-balances the training set, which partly replaces the need for WeightedRandomSampler, but we kept both for completeness and correctness.
- EfficientNet comparison: not yet tested due to time constraints. Planned for Phase 3 if GPU access is obtained via Google Colab.

3.4 Planned Next Steps for Phase 3

- Train on full dataset using Google Colab GPU for 20–30 epochs.
- Test with higher input resolution (224×224) to capture finer aerial details.
- Experiment with Focal Loss as an alternative to weighted cross-entropy.
- Apply targeted augmentation specifically for Sedan/SUV/Van to reduce their confusion.
- Generate predictions on the official unlabelled validation set (770 images).

4. References

[1] S. Low et al., "Multi-modal Aerial View Object Classification Challenge Results - PBVS 2022," CVPRW 2022.

[2] https://codalab.lisn.upsaclay.fr/competitions/1392

[3] J. M. Johnson and T. M. Khoshgoftaar, "Survey on deep learning with class imbalance," Journal of Big Data, 2019.

[4] https://towardsai.net/p/l/multi-class-model-evaluation-with-confusion-matrix-and-classification-report